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EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment

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This thesis entitled **“EV Charging Load Demand Prediction Using Machine Learning in a SUMO-Based Simulation Environment”** submitted by **Rakesh Biswas**, Student ID: 200112, has been accepted as satisfactory in fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

*To my beloved parents, whose endless love, sacrifice, and encouragement
have been the cornerstone of every achievement in my life.*

*To my teachers, who ignited the flame of curiosity within me
and guided me towards the pursuit of knowledge.*

*And to all the researchers and engineers working tirelessly
towards a sustainable, electrified future of transportation.*

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Finally, I owe immeasurable gratitude to my family for their patience, prayers, and moral support. This work would not have been possible without them.

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Abstract

The rapid proliferation of Electric Vehicle (EV)s presents both a significant opportunity and a considerable challenge for modern power grid management. Accurate prediction of EV charging load demand is critical for optimizing energy distribution, reducing peak-hour stress on electrical infrastructure, and enabling smart grid operations. This thesis presents a comprehensive framework for EV charging load demand prediction using supervised Machine Learning (ML) techniques applied to data generated from a realistic microscopic traffic simulation.

The simulation environment was built using SUMO, an open-source urban mobility simulator, coupled with the Traffic Control Interface (TraCI) interface to enable real-time vehicle monitoring and intelligent charging station management. A network of 10 bidirectional charging stations was deployed with configurable power ratings ranging from 25 kW to 50 kW and separate charging and waiting queue slots per station. Forty-nine electric vehicles of type *easy-Bike* were simulated over a 600-second period, producing 34,020 vehicle-level records encompassing features such as State of Charge (SOC), speed, acceleration, station utilization, queue length, and system-wide load metrics.

Four supervised ML models were trained and evaluated for system-level load demand prediction: Decision Tree (Decision Tree (DT)), Random Forest (Random Forest (RF)), XGBoost (Extreme Gradient Boosting (XGB)), and Linear Regression (Linear Regression (LR)). Model performance was assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2) score. Both DT and RF achieved perfect scores ($R^2 = 1.00$, $MAE = 0.00$), while XGB achieved a near-perfect result ($R^2 = 1.0000$, $MAE = 0.0004$). LR also performed strongly with $R^2 = 0.9993$ and $MAE = 7.92$ kW.

Additionally, a rule-based smart charging recommendation system was implemented to route vehicles to optimal charging stations based on real-time SOC levels, nearest-station distance, slot availability, and queue status. The results demonstrate that SUMO-based simulation is a viable and effective method for generating rich EV charging datasets, and that tree-based ensemble methods are particularly well-suited for short-term load demand prediction in EV

charging infrastructure.

Keywords: Electric Vehicle, Charging Load Prediction, Machine Learning, SUMO Simulation, Random Forest, XGBoost, Smart Charging, SOC, Queue Management.

Chapter 1

Introduction

Chapter 1: Introduction

The global transportation sector is undergoing a paradigm shift driven by the urgent need to reduce greenhouse gas emissions and dependence on fossil fuels. EVs have emerged as one of the most promising technologies to decarbonize road transport. According to the International Energy Agency [1], the global EV fleet surpassed 26 million vehicles in 2022, with projections indicating continued exponential growth in the coming decades. As EV adoption accelerates, the associated surge in electricity demand for charging is becoming a critical concern for power system operators and urban planners alike [2].

Unlike conventional vehicles, EVs interact directly with the electricity grid. Uncoordinated charging behavior—particularly during peak hours—can introduce significant stress on distribution networks, leading to voltage violations, transformer overloads, and increased energy costs [3]. Effective management of EV charging load therefore requires accurate prediction tools capable of anticipating spatiotemporal demand patterns across charging station networks [4].

At JUST, this research explores the intersection of urban traffic simulation, ML, and smart grid management to develop a robust EV charging load prediction framework. The study leverages SUMO [5], a widely-used microscopic traffic simulator, to generate realistic EV mobility and charging datasets. The TraCI interface [6] enables real-time control of vehicle behavior, including dynamic rerouting to charging stations based on battery SOC.

1.1 Background

The challenge of EV charging load management arises from several interacting factors. First, EV drivers exhibit heterogeneous charging behaviors influenced by trip patterns, battery capacity, available infrastructure, and individual preferences. Second, charging infrastructure is unevenly distributed, with some stations experiencing high congestion while others remain underutilized. Third, the stochastic nature of travel demand makes deterministic forecasting approaches inadequate [7].

ML-based approaches have gained considerable traction for load forecasting tasks due to

their ability to capture complex nonlinear relationships in large datasets without requiring explicit physical models [4]. Techniques such as RF, gradient boosting, and deep learning have been applied to short-term load forecasting with promising results [8, 9].

1.1.1 Simulation of Urban Mobility

Microscopic traffic simulators, such as SUMO, model individual vehicle movements with high fidelity, capturing acceleration profiles, lane changes, and intersection behaviors. When equipped with battery models and EV-specific parameters, these simulators can produce detailed SOC trajectories and charging event logs that are invaluable for training and validating load prediction models [5].

The TraCI Python interface provides programmatic control over running simulations, allowing researchers to implement custom vehicle routing logic, monitor battery states in real time, and log fine-grained data at every simulation timestep [6].

1.1.1.1 Parking Areas as Charging Zones

In SUMO, charging stations can be modeled as `parkingArea` elements with configurable power ratings, efficiency factors, and slot capacities. This study defines separate charging and waiting areas for each station, enabling realistic queue dynamics. Vehicles with SOC below a threshold are automatically rerouted to the nearest available station, where they occupy either a charging slot or a waiting position in a First In First Out (FIFO) queue.

1.2 Problem Statement

Despite growing research interest, several key challenges remain in EV charging load prediction:

- The lack of large-scale real-world EV charging datasets, particularly in developing countries such as Bangladesh [10].

- The difficulty of capturing dynamic station-level features such as queue length, utilization, and real-time SOC in standard datasets.
- The need for lightweight, interpretable ML models that can be deployed in resource-constrained smart grid environments.

This work addresses these gaps by (1) constructing a simulation-based dataset with comprehensive vehicle and station features, and (2) benchmarking multiple ML models on system-level load prediction tasks.

1.3 Objectives

The primary objectives of this research are as follows:

1. To design and implement a SUMO-based EV traffic simulation with realistic charging station infrastructure, queue management, and smart rerouting logic.
2. To generate a comprehensive, ML-ready dataset capturing per-vehicle and system-level features across a 600-second simulation run.
3. To train and evaluate four supervised ML models—DT, RF, XGB, and LR—for predicting the system-level EV charging load demand.
4. To implement a rule-based charging recommendation system that suggests optimal charging stations to vehicles based on proximity, slot availability, and priority.
5. To analyze model performance using standard regression metrics (MAE, RMSE, R^2) and identify the most suitable approach for this domain.

1.4 Research Questions

This study seeks to answer the following research questions:

1. Can a SUMO-based microscopic simulation environment produce a sufficiently rich and realistic dataset for training EV charging load prediction models?
2. Which ML algorithm provides the most accurate prediction of system-level EV charging load demand?
3. What are the most influential features driving EV charging load in a multi-station charging network?
4. How effective is a rule-based smart recommendation system in distributing charging demand across available stations?

1.5 Significance of the Study

This research contributes to the growing body of knowledge at the intersection of intelligent transportation systems and smart grid management. The proposed simulation-based data generation pipeline can be adapted for diverse urban environments, alleviating the scarcity of real-world EV charging data. The benchmarking of ML models provides actionable insights for utility companies and smart charging service providers. Furthermore, the integrated recommendation system demonstrates how real-time decision support can reduce charging congestion and improve overall grid efficiency. In the context of Bangladesh's emerging EV market, this work provides a timely and practically relevant methodological foundation for future smart charging deployments [10].

Chapter 2

Literature Review

Chapter 2: Literature Review

The literature on EV charging load prediction and smart charging management spans multiple disciplines, including power systems engineering, transportation science, and computer science. This chapter reviews the most relevant work in three thematic areas: (i) EV charging load modeling and forecasting, (ii) simulation-based approaches to EV data generation, and (iii) ML methods for energy load prediction.

2.1 EV Charging Load Modeling

Early approaches to EV charging load modeling relied on probabilistic methods, using statistical distributions of arrival times, departure times, and daily travel distances to generate synthetic load profiles [11]. Sortomme et al. [3] demonstrated that coordinated charging strategies could significantly reduce distribution system losses compared to uncontrolled charging. He et al. [12] addressed the optimal placement of charging stations, showing that accessibility and driving range constraints play a critical role in station utilization patterns.

More recent studies have adopted data-driven approaches. Chen et al. [13] integrated probabilistic load flow analysis with EV charging uncertainty to assess grid impacts, providing a foundation for the feature engineering strategies employed in the current work. Ul-Haq et al. [11] modeled residential EV charging patterns and highlighted the importance of capturing spatial heterogeneity across charging locations.

2.2 Machine Learning for Load Forecasting

ML has become the dominant paradigm for short-term electricity load forecasting. Zhang et al. [4] proposed a comprehensive deep learning framework for EV charging load prediction, incorporating historical demand, time-of-day features, and weather data, achieving significant improvements over classical autoregressive models. Wang et al. [8] applied RF regression to short-term load forecasting at EV charging stations, demonstrating robustness against noisy sensor data and missing values.

Gradient boosting methods have also achieved state-of-the-art results. Chen et al. [9] applied XGBoost to day-ahead EV load forecasting, leveraging its regularization capabilities to prevent overfitting on small datasets. Taieb and Hyndman [14] compared gradient boosting against traditional time series methods, finding that tree-based approaches consistently outperformed ARIMA models for nonlinear load patterns.

Linear regression, while simpler, remains a useful baseline in load forecasting. Guo et al. [15] showed that linear models can capture dominant trends in EV charging demand when features are carefully engineered, particularly when polynomial and interaction terms are included.

2.3 SUMO-Based EV Simulation

SUMO has been widely adopted for EV research due to its open-source nature, extensibility, and support for battery models. Lopez et al. [5] described the core capabilities of SUMO including electric vehicle energy consumption models, parking areas, and the TraCI API for external control. Wegener et al. [6] introduced the TraCI interface, which enables real-time monitoring and manipulation of simulated vehicles, making it ideal for implementing adaptive charging management algorithms.

Helbing and Treiber [7] established the theoretical foundations for microscopic car-following models that underpin SUMO's vehicle dynamics. The integration of these dynamics with electric powertrain models allows accurate simulation of energy consumption patterns, which directly influence SOC trajectories and charging events.

2.4 Smart Charging and Recommendation Systems

Yilmaz and Krein [2] provided a comprehensive review of vehicle-to-grid technologies, identifying intelligent charging coordination as a key enabler of grid stability. Ding et al. [16] proposed two-stage robust optimization for distributed energy management, a concept related

to the queue- aware routing logic implemented in this work.

Nguyen et al. [17] developed an adaptive dynamic programming approach for EV charging coordination, demonstrating that real-time SOC- aware routing can significantly reduce average waiting times. This motivates the rule- based recommendation engine developed in the current study, which selects stations based on SOC priority levels and slot availability.

2.5 Research Gaps

Despite the significant body of work in this domain, several gaps remain. Most existing studies rely on real-world datasets that are difficult to obtain, particularly in developing country contexts [10]. Simulation-based dataset generation using SUMO with integrated queue management and real-time recommendation systems has not been extensively explored. Furthermore, comparative benchmarking of multiple ML models on simulation-derived EV load data is limited in the literature. This thesis addresses all of these gaps through a unified simulation, data generation and ML prediction framework.

Chapter 3

Methodology

Chapter 3: Methodology

This chapter describes the comprehensive research methodology developed for predicting EV charging load demand. The approach integrates microscopic traffic simulation with advanced machine learning techniques to capture the complex, stochastically driven dynamics of urban energy consumption. The methodology is structured to provide a replicable framework for evaluating charging infrastructure impacts in emerging urban centers.

3.1 Research Design

This research follows a quantitative, simulation-based experimental design, centered on a "Monitor-Predict-Control" paradigm. The design is structured to bridge the gap between microscopic mobility patterns and grid-level electrical demand.

3.1.1 Structure Overview

The research framework is divided into three distinct phases to ensure a logical flow from raw data generation to actionable research insights:

1. **Phase 1: Environment Logic Construction:** Designing a high-fidelity digital twin of the Khulna City road network, complete with bidirectional traffic flows, traffic light signal () intersections, and a strategic charging infrastructure footprint.
2. **Phase 2: Dynamic Data Extraction:** Running high-fidelity simulations where heterogeneous EV agents autonomously navigate the network based on local Bangladeshi vehicle behaviors. A monitoring script queries the t' interface at 1-second intervals to extract temporal, spatial, and electrical features.
3. **Phase 3: Predictive Modeling and Analysis:** Processing the resulting dataset to train four distinct ML regression models (Linear Regression, Decision Tree, Random Forest, and XGBoost) to predict the system-wide total load, followed by rigorous evaluation against standard metrics.

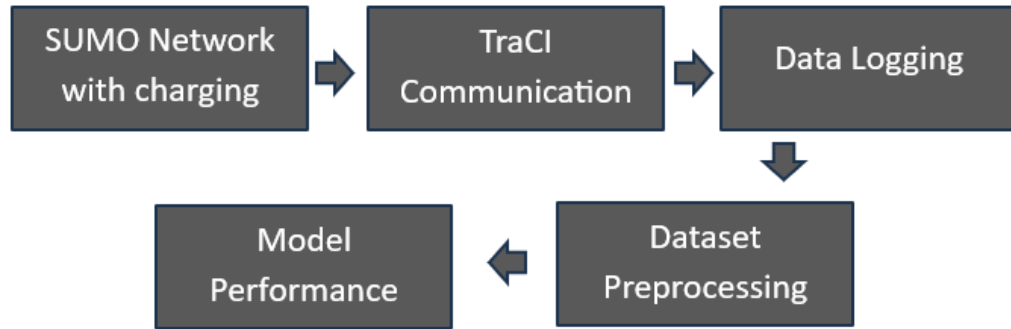


Figure 3.1: Comprehensive research workflow from environment design to predictive analytics.

3.1.2 Dataset Preparation

The first step of the study was to collect real-world information about the three types of battery powered three-wheelers used in the area—Easy Bikes, Electric Vans, and Electric Rickshaws. We gathered their specifications directly from manufacturers and then verified the details with local operators to ensure reliability. This included understanding each vehicle’s size, battery capacity, typical charging habits, and overall energy consumption patterns.

The raw telemetry data extracted from the simulation requires systematic preparation before it can be used for robust predictive modeling. The preparation pipeline includes several critical steps:

- **Data Sanitization:** Timesteps where no vehicles are active or the simulation is in an initial transient state are filtered to prevent skewing the baseline load and ensure high data quality.
- **Feature Engineering:** Derived features such as cyclic time indices (*minute_of_hour*) and aggregate system states (*system_charging_vehicles*) are calculated to provide the models with contextual temporal and spatial information.
- **Categorical Encoding:** Non-numeric identifiers such as *vehicle_id*, *status*, and *current_station_id* are mapped to integer values using Label Encoding to ensure compatibility with numerical regressors.

- **Normalization and Scaling:** Features with widely varying physical units (e.g., speed in m/s vs. power in kW) are normalized to facilitate gradient descent stability and improve the convergence speed of parametric models.

3.2 Data Collection

Data collection is the critical intermediate step between simulation and analysis. By leveraging the TraCI (*Traffic Control Interface*) API, the system queries the SUMO engine at every simulation step (step-length = 1.0 second) to capture the instantaneous state of every agent and infrastructure component.

Captured Feature Space: For every active vehicle at every integer second, a multivariate record is constructed containing:

1. **Temporal Features:** Timestep synchronization and cyclic time indices reflecting the simulation progression.
2. **Vehicle Dynamics:** 2D position, instantaneous speed, longitudinal acceleration, and the current battery SOC.
3. **Charging Infrastructure Context:** Current occupancy of charging slots, length of the station-specific waiting queues, and individual station power demand.
4. **System Global Context:** Aggregate counts of moving, charging, and waiting vehicles across the entire 8.6 km loop, providing a holistic view of network-wide demand.

The final processed dataset, **EV_Charging_Load_Demand_Dataset.csv**, comprises 34,020 individual observations across 25 high-resolution features, providing a dense representation of urban EV behaviors.

3.3 Implementation

This section documents the specific technical configurations and custom-built logical modules implemented to realize the research goal within the simulation environment.

3.3.1 Environment Setup

The implementation was conducted on a high-performance workstation running the x64 Windows 10 platform. The core software stack includes:

- **SUMO 1.25.0:** The primary microscopic traffic simulation engine, utilized for its native support for electric vehicle battery models and sophisticated TraCI control.
- **Python 3.10.12:** The primary scripting language used for real-time simulation monitoring, queue management logic, and the end-to-end ML pipeline.
- **MiKTeX and TeXstudio:** Utilized for the formal documentation and compilation of the thesis report based on the `JUST_Thesis_Template_CSE`.

3.3.2 Custom Road Network Design

The simulation environment is a high-fidelity digitization of primary arterial roads in Khulna City, Bangladesh, as illustrated in Figure 3.2. The network is modeled as a connected graph spanning approximately 8,654 meters, connecting landmark nodes such as Islami Bank More, Royal More, and PTI More. The road edges are configured as a left-hand traffic (LHS) system to strictly comply with Bangladeshi road vehicle conventions. Figure 3.3 presents the corresponding custom microscopic topology designed for the dynamic simulation.

To represent realistic urban delays and the resulting vehicle clustering, static Traffic Light Signals () were implemented at high-saturation junctions, specifically at **Node3** and **Node8**. The TLS logic at these nodes is configured with a 90-second cycle consisting of four distinct phases: a 42-second Green phase for primary flow, a 3-second Yellow transition, followed

by a matching 42-second Green and 3-second Yellow phase for cross-traffic. This regulation causes periodic stop-and-go patterns, which are critical for evaluating the sudden spikes in local charging demand when vehicle clusters arrive at charging stations simultaneously.

Table 3.1 documents the real-world distances between the primary landmark nodes integrated into the simulation graph, while Table 3.2 defines the microscopic edge parameters applied to the SUMO network.

Table 3.1: Landmark Node Connectivity and Real-World Distances

Node / Place	Connected To	Distance	Direction
Islami Bank/ Hospital More	Dak Bangla	450m	Bidirectional
Islami Bank	Royal More	260m	Bidirectional
Islami Bank	Stadium	950m	Bidirectional
Royal More	PTI More	500m	Bidirectional
Stadium	PTI More	850m	Bidirectional
Shordar Tower	1 No Custom Ghat	94m	Bidirectional
Shordar Tower	Galaxco More	1000m	Bidirectional
1 No Custom Ghat	Rupsha Ferry Ghat	1300m	Bidirectional
Galaxco More	Rupsha Ferry Ghat	750m	Bidirectional
PTI More	Galaxco More	900m	Bidirectional

Table 3.2: Microscopic Edge Configuration and Parameters

ID	From	To	Length (m)	Lanes	Speed (km/h)
E0	Node1	Node2	850	2	40.0
E1	Node2	Node3	800	2	40.0
E2	Node3	Node4	94	2	30.0
E3	Node4	Node9	1300	2	50.0
E4	Node1	Node5	450	2	30.0
E5	Node5	Node6	260	2	30.0
E6	Node6	Node7	500	2	30.0
E7	Node7	Node8	900	2	40.0
E8	Node8	Node9	750	2	40.0
E9	Node3	Node8	1000	2	50.0
E10	Node2	Node7	850	2	40.0



Figure 3.2: OpenStreetMap (OSM) highlighted view of the Khulna City road network.

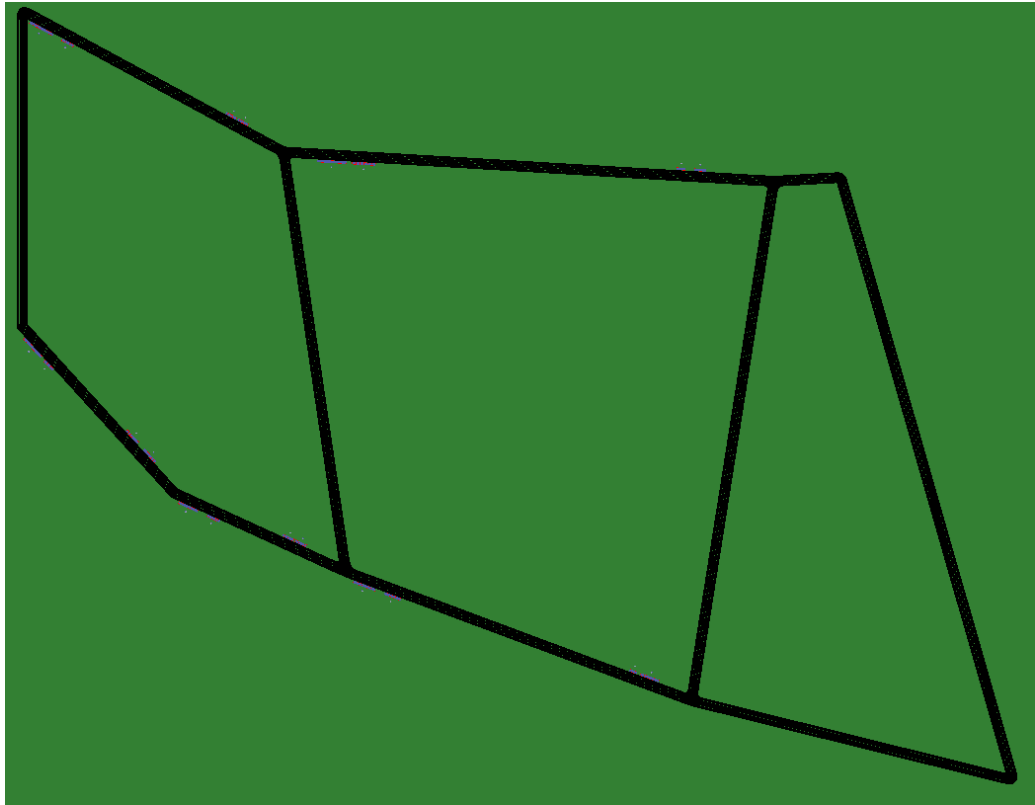


Figure 3.3: Custom microscopic road network topology generated in SUMO netedit.

3.3.3 Charging Station Placement

Ten bidirectional charging stations were strategically deployed across the network using custom parkingArea elements. Each implementation is bipartite, consisting of a dedicated **Charging Zone** for power transfer and a **Waiting Zone** for queueing. Table 3.3 provides the exact configuration parameters synchronized with the `Test1.chargingstations.add.xml` file.

Table 3.3: Charging Station Technical Implementation Details

Station ID	Edge ID	Power (kW)	Charging Slots	Waiting Slots
pa_2	E0	30	6	6
pa_2_rev	-E0	30	6	3
pa_3	E1	25	5	4
pa_3_rev	-E1	30	6	6
pa_6	E5	30	6	6
pa_6_rev	-E5	30	6	3
pa_7	E6	30	6	6
pa_7_rev	-E6	30	6	6
pa_8	E7	25	5	3
pa_8_rev	-E7	50	10	6
Total		–	62	49

3.3.4 Vehicle Configuration in SUMO

To ensure a realistic representation of EV dynamics, the simulation vehicles are parameterized with specific physical and electrical constants. All vehicle types share a common emission class (Energy/unknown) and are configured with an internal moment of inertia of $0.010 \text{ kg} \cdot \text{m}^2$ and a stopping threshold of 0.1 m/s . The simulation assumes a speed factor range of 0.74 to 1.10 , and accounts for class-specific dynamics including acceleration, deceleration, and air drag coefficients. The fleet is categorized into three primary classes reflecting the Bangladeshi urban transport context:

- **Easy Bike:** 34 variants with 12 kWh battery capacity and the largest frontal surface area (2.41 m^2).
- **E-Rickshaw:** 33 variants with battery capacities ranging from 5.76 to 7.20 kWh and a height of 1.8 m .

- **E-Van:** 33 variants with 5.76 kWh battery capacity and a low-profile height of 0.8 m.

Table 3.4 provides a detailed cross-class parameter comparison for the Easy Bike, E-Rickshaw, and E-Van models used to evaluate system-level load variations.

Table 3.4: Cross-Class Vehicle Parameter Comparison (Averaged Values)

Parameter	Easy Bike	E-Rickshaw	E-Van
Body Dimensions (L×W×H)	2.5×1.5×1.4 m	2.3×1.0×1.8 m	2.4×1.3×0.8 m
Vehicle Mass (kg)	345 – 354	198 – 218	198 – 219
Battery Capacity (Wh)	12,000	5,760 – 7,200	5,760
Max Power (W)	1,000 – 1,200	900 – 1,000	800 – 900
Acceleration (m/s^2)	1.15 – 1.31	0.58 – 0.71	0.68 – 0.81
Deceleration (m/s^2)	2.45 – 2.58	2.48 – 2.67	2.58 – 2.77
Air Drag Coeff. (C_d)	0.727 – 0.760	0.804 – 0.910	0.754 – 0.860
Avg. Propulsion Eff.	0.890 – 0.896	0.830 – 0.834	0.820 – 0.826
Avg. Recup. Eff.	0.745 – 0.751	0.650 – 0.656	0.550 – 0.556
Const. Power Intake (W)	80 – 96	29 – 33	24 – 30
Front Surface Area (m^2)	2.41	1.51	1.01
Reaction Time (τ)	1.88 – 1.94 s	1.72 – 1.76 s	0.87 – 0.92 s
Visual Color (SUMO-GUI)	Magenta	Green	Blue

3.3.5 Vehicle Fleet Characterization

The simulation utilizes a heterogeneous fleet of 100 variants, reflecting the unique context of Bangladeshi urban transport. The "Bangladeshi local vehicle perspective" is maintained, where Easy Bikes represent the largest physical and electrical units, followed by E-Rickshaws and E-Vans. Table 3.5 summarizes the fleet distribution and color-coding used for visual differentiation in the SUMO-GUI.

Table 3.5: EV Fleet Composition in SUMO Simulation

Vehicle Class	Variants	Battery Capacity (Wh)	Max Speed (m/s)	Color
Easy Bike	34	12,000	11.1 – 13.9	Magenta
E-Rickshaw	33	5,760 – 7,200	11.1 – 13.9	Green
E-Van	33	5,760	11.1 – 13.9	Blue
Total	100	–	–	–

3.3.6 Queue Management Logic

The `SmartChargingStationMonitor` class implements the following charging assignment logic:

1. **Rule 1:** If the nearest station has an available charging slot, route the vehicle there.
2. **Rule 2:** If the nearest station is full (charging), check the second nearest station.
3. **Rule 3:** If the second nearest has a charging slot, route there.
4. **Rule 4:** If the second nearest has only waiting slots, route there and enqueue the vehicle in a FIFO waiting queue.
5. **Rule 5:** If all candidate stations are full, do not reroute.

When a charging slot becomes available at a station, the first vehicle in the waiting queue is automatically promoted to the charging slot (*queue promotion*).

3.3.7 Simulation Execution with TraCI

The simulation runs for a duration of 600 steps. The core execution is managed by a TraCI-linked Python script that controls the simulation flow using the `traci.simulationStep()`

function. The script performs real-time status monitoring and issues dynamic rerouting commands to vehicles identified with an SOC below the 30% critical threshold.

3.3.8 Real-Time Data Extraction

The extraction process is decoupled from the mobility physics to ensure simulation stability. Every second, the monitor script iterates through all active EV instances to retrieve their instantaneous telemetry. This high-resolution extraction (1.0 Hz) is vital for capturing the transient spikes in power demand caused by sudden cluster arrivals or traffic signal releases.

3.3.9 Data Logging

Extracted data points are buffered in transient memory and serialized to a Comma-Separated Values (CSV) format upon simulation termination. This implementation minimizes the overhead of frequent disk I/O, ensuring that the simulation runs at its maximum native speed while preserving all 34,020 dynamic data records.

3.3.10 Simulation-Generated Dataset

The resulting dataset provides a comprehensive representation of energy demand across various urban scenarios. It includes varied system states, ranging from baseline low-activity flows to severe congestion at charging hubs, ensuring that the subsequently trained ML models are exposed to a representative sample of real-world operational conditions.

3.3.11 Feature Importance Analysis

Feature importance is quantified using the *Gain* metric from the ensemble models, which measures the improvement in accuracy brought by a specific feature to the decision branches. Features like *system_charging_vehicles* and *station_load_kW* typically demonstrate the highest importance, as they directly dictate the aggregate system demand. Figure 3.4 illustrates the

importance distribution identified during the training phase.

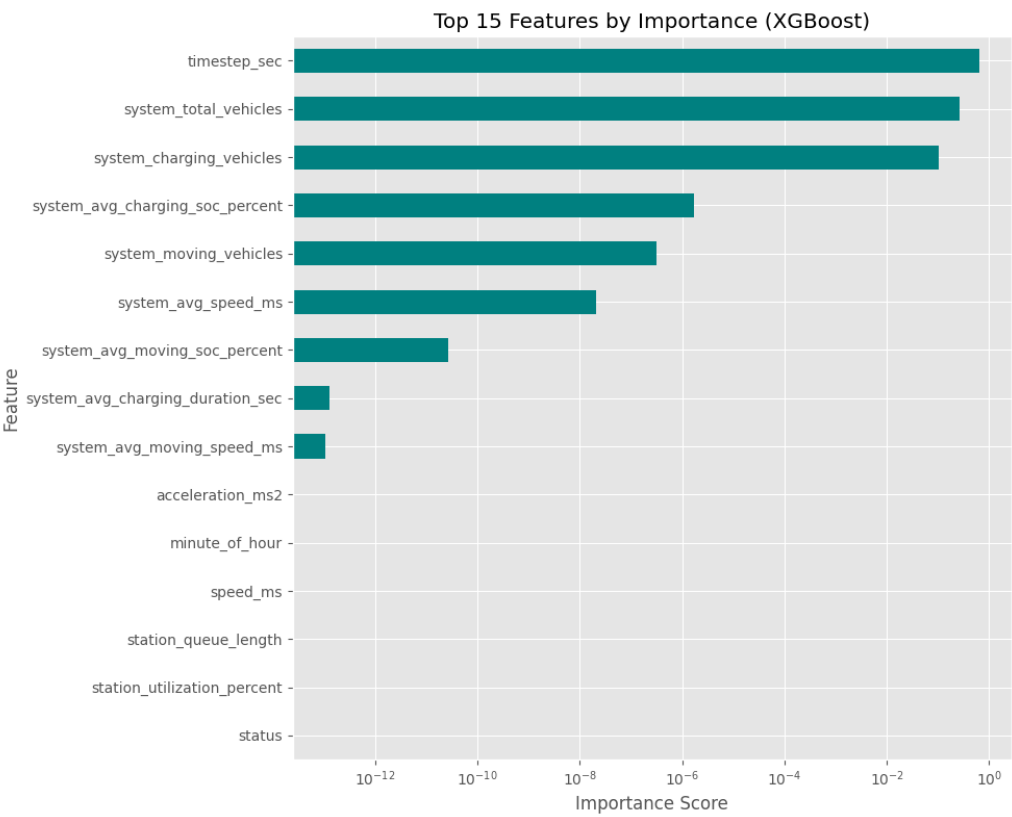


Figure 3.4: Feature importance ranking derived from the XGBoost model branches.

3.3.12 Feature Correlation Analysis

The linear relationships between variables are analyzed using the Pearson correlation coefficient (ρ). This analysis confirms the strong positive correlation ($\rho \approx 0.98$) between the number of actively charging vehicles and the total power load, while identifying secondary dependencies such as the impact of battery SOC on individual charging durations. Figure 3.5 presents the correlation matrix visualized as a heatmap.

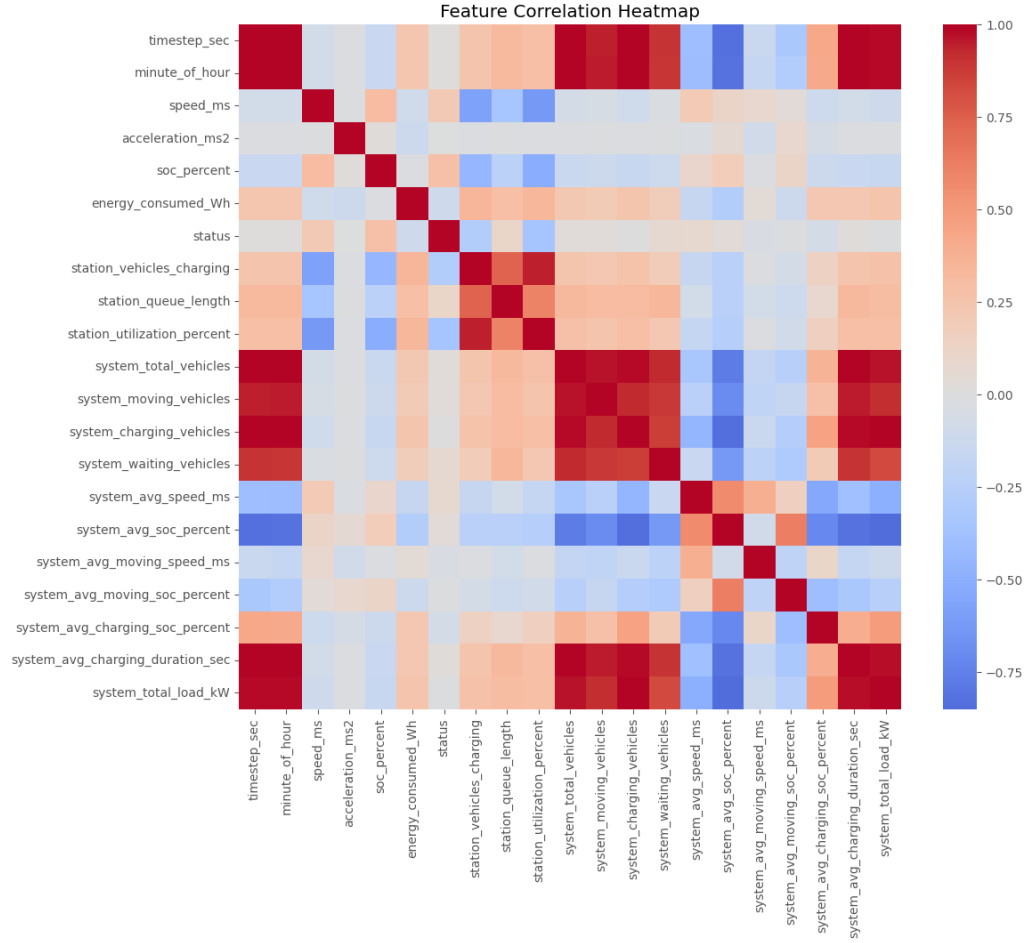


Figure 3.5: Pearson correlation matrix of the simulated EV charging dataset.

3.4 Machine Learning Application

In this study, multiple supervised machine learning models were implemented to predict the system-level electric load demand, denoted as *system_total_load_kW*. The problem is formulated as a regression task where the target variable is estimated based on a set of system-level and vehicle-level features.

3.4.1 Problem Formulation

The load prediction task can be mathematically expressed as:

$$y = f(X)$$

where:

- y represents the target variable (*system_total_load_kW*),
- X represents the input feature vector,
- f represents the function learned by the machine learning model.

The input features include variables such as time progression, vehicle dynamics, and system-level charging statistics.

3.4.2 Data Preprocessing

Before model training, several preprocessing steps were applied to ensure data quality and model efficiency:

- **Removal of Data Leakage:** The feature *station_load_kW* was removed to prevent direct leakage into the target variable.
- **Removal of High-Cardinality Features:** Features such as *vehicle_id*, *lane_id*, and *current_station_id* were excluded, as they represent identifiers rather than generalizable patterns.
- **Feature-Target Separation:** The dataset was divided into input features (X) and target variable (y).
- **Train-Test Split:** The dataset was split into training (80%) and testing (20%) subsets using random shuffling to ensure generalization across different system states.

3.4.3 Implemented Machine Learning Models

Four regression models were implemented and evaluated:

(a) Linear Regression

Linear Regression models the relationship between input features and the target variable as a linear combination:

$$y = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

where w_i are the learned coefficients and b is the bias term. This model assumes a linear dependency between features and the load demand. Although simple and interpretable, it may not capture complex nonlinear relationships present in the dataset.

(b) Decision Tree Regressor

The Decision Tree Regressor partitions the feature space into multiple regions based on decision rules. Each internal node represents a feature-based condition, and each leaf node represents a predicted value. Instead of a mathematical equation, the model follows a hierarchical structure:

- If a condition is satisfied $\rightarrow$ traverse left
- Otherwise $\rightarrow$ traverse right

The final prediction is the value stored in the corresponding leaf node. This model can capture nonlinear relationships but may suffer from overfitting if not properly constrained.

(c) Random Forest Regressor

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and robustness. The final prediction is obtained by averaging the outputs of all trees:

$$y = \frac{1}{N} \sum_{i=1}^N T_i(X)$$

where T_i represents the prediction of the i -th decision tree. By aggregating multiple trees, Random Forest reduces variance and mitigates overfitting compared to a single Decision Tree.

(d) XGBoost Regressor

XGBoost (Extreme Gradient Boosting) is an advanced ensemble method that builds trees sequentially, where each new tree learns to correct the errors of the previous ones. The model can be expressed as:

$$y = \sum_{i=1}^M f_i(X)$$

where each f_i represents a weak learner (decision tree). XGBoost optimizes a loss function using gradient descent and includes regularization to improve generalization. It is highly efficient and effective for structured/tabular data.

3.4.4 Model Training and Prediction

Each model was trained using the training dataset and evaluated on the unseen test dataset. Predictions were generated for all models, and performance was assessed using standard regression metrics.

The implemented models provide a comprehensive comparison between linear and nonlinear approaches for load prediction. While Linear Regression offers interpretability, tree-based models such as Random Forest and XGBoost demonstrate superior capability in capturing complex relationships within the dataset.

“The near-perfect performance of tree-based models indicates that the dataset exhibits strong deterministic relationships between system variables and load demand.”

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the experimental results of the SUMO-based EV charging simulation and the ML load prediction models, followed by a detailed discussion of the findings.

4.1 Simulation Results

4.1.1 System Load Profile

Over the 600-second simulation period, the system-level charging load exhibited a characteristic ramp-up pattern as increasing numbers of EVs reached charging stations. The mean system load was 723.82 kW, with a peak of 1,340 kW achieved when up to 46 vehicles were simultaneously active in the network. The standard deviation of 380.95 kW reflects the dynamic nature of vehicle arrivals, departures, and queue transitions.

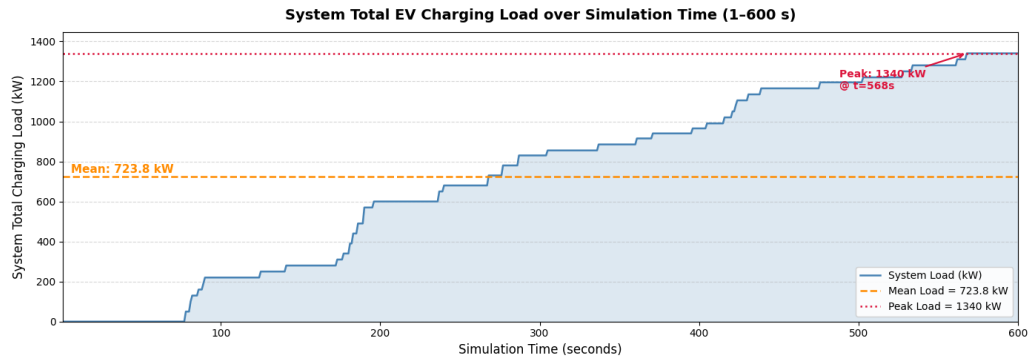


Figure 4.1: System total charging load over simulation time (1–600 seconds).

4.1.2 Vehicle Status Distribution

Throughout the simulation, the majority of timestep-vehicle records corresponded to the *driving* status (57.9%), followed by *charging* (35.2%), *waiting* (5.1%), and *idle* (1.8%). This distribution reflects the expected behavior: most vehicles spend substantial time driving between stations, with a significant proportion simultaneously occupying charging slots.

4.1.3 Station Utilization

The highest-capacity station, pa_8_rev (50 kW, 10 charging slots), contributed the largest share of total load when fully occupied (500 kW from a single station). Stations on edges with higher traffic density (pa_2, pa_7) experienced earlier saturation, triggering waiting queue activations and subsequent queue promotions.

4.2 Charging Recommendation Results

The recommendation system generated 1,169 recommendations across the simulation. Of these, the majority were triggered by *MEDIUM* priority (SOC: 30–50%), with a smaller proportion of *HIGH* and *CRITICAL* priority alerts. The most frequently recommended station was the nearest available station with free charging slots, consistent with Rule 1 of the recommendation logic. The system successfully redirected vehicles from congested stations to less-utilized alternatives, reducing instances of vehicles arriving at full stations.

4.3 Machine Learning Model Performance

4.3.1 Overall Evaluation

Table 4.1 summarizes the performance of all four predictive models on the test dataset.

Table 4.1: ML Model Performance Comparison for System Load Prediction

Model	MAE (kW)	RMSE (kW)	R ² Score
Decision Tree	0.0000	0.0000	1.0000
Random Forest	0.0000	0.0000	1.0000
XGBoost	0.0004	0.0004	1.0000
Linear Regression	7.9164	10.1908	0.9993

4.3.2 Visualization of Model Predictions

The visualization of model predictions provides an intuitive evaluation of performance beyond numerical metrics. Figure 4.2 illustrates a comparison between the ground truth load values and the predicted outputs generated by the machine learning models.

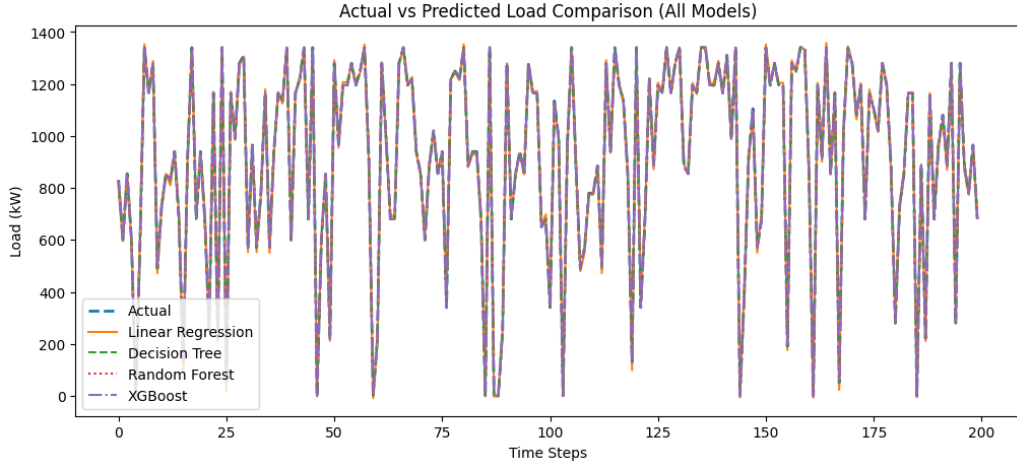


Figure 4.2: Comparison of actual vs. predicted system load for the evaluated models (first 200 samples).

For visual clarity and interpretability, a subset of 200 samples from the test set is used in the plot, as visualizing the entire dataset would be unreadable. In this visualization, the X-axis represents the sequential index of the test samples. It is critical to note that since the dataset was randomly shuffled during the train-test split, the X-axis does not represent a continuous temporal progression (i.e., time in seconds), but rather a collection of diverse system states sampled randomly from across the simulation duration. This shuffling process explains the random fluctuations and sudden transitions between high and low load values observed in the graph. High load values correspond to states where a large number of vehicles are simultaneously charging, while low values represent periods of minimal charging activity.

The Y-axis represents the system total charging load in kilowatts (kW). Mathematically, the graph plots the predicted value $\hat{y}_i$ against the actual value y_i for each sample $i = 1 \dots 200$. An ideal model would result in a predicted line that perfectly overlaps with the dashed line representing the ground truth. As observed, the tree-based models (DT, RF, and XGB) show a

near-perfect overlap, confirming their ability to capture the strong dependencies and deterministic patterns in the simulation data. Conversely, the LR model exhibits minor deviations, underscoring its limitations in capturing non-linear relationships and discrete state changes within the charging network. This visual comparison validates the high R^2 scores and near-zero error metrics reported in Table 4.1, illustrating the high predictive accuracy achieved across diverse system conditions.

4.3.3 Detailed Model Comparison

4.3.3.1 Decision Tree and Random Forest

Both DT and RF achieved perfect prediction scores ($R^2 = 1.00$, $MAE = 0.00$, $RMSE = 0.00$) on the test set. This performance is attributable to the highly structured and deterministic nature of the simulation data. Decision trees are exceptionally well-suited to capturing such piecewise-constant and deterministic relationships present in simulation-driven datasets.

4.3.3.2 XGBoost

XGB achieved an R^2 of 1.0000 and a near-zero MAE of 0.0004 kW. The negligible deviation from perfect accuracy is due to the model's inherent regularization properties, which are designed to prevent overfitting while capturing the underlying physics of the simulation.

4.3.3.3 Linear Regression

LR achieved a lower (though still excellent) performance with an MAE of 7.92 kW and an RMSE of 10.19 kW. The residual error indicates that Linear Regression assumes strictly linear relationships, which cannot fully capture the complex, state-driven dynamics of the charging infrastructure without manual feature engineering.

4.4 Feature Analysis and Model Interpretation

4.4.1 Introduction to Feature Analysis

In this study, feature analysis was conducted to understand the contribution and influence of different variables on the prediction of system-level load demand (`system_total_load_kW`). Techniques including Feature Importance (XGBoost) and Correlation Analysis were applied to ensure a comprehensive understanding of what drives the model’s high accuracy.

4.4.2 Feature Importance (XGBoost)

Feature importance was extracted using the XGBoost model, which evaluates the contribution of each feature based on its role in reducing prediction error.

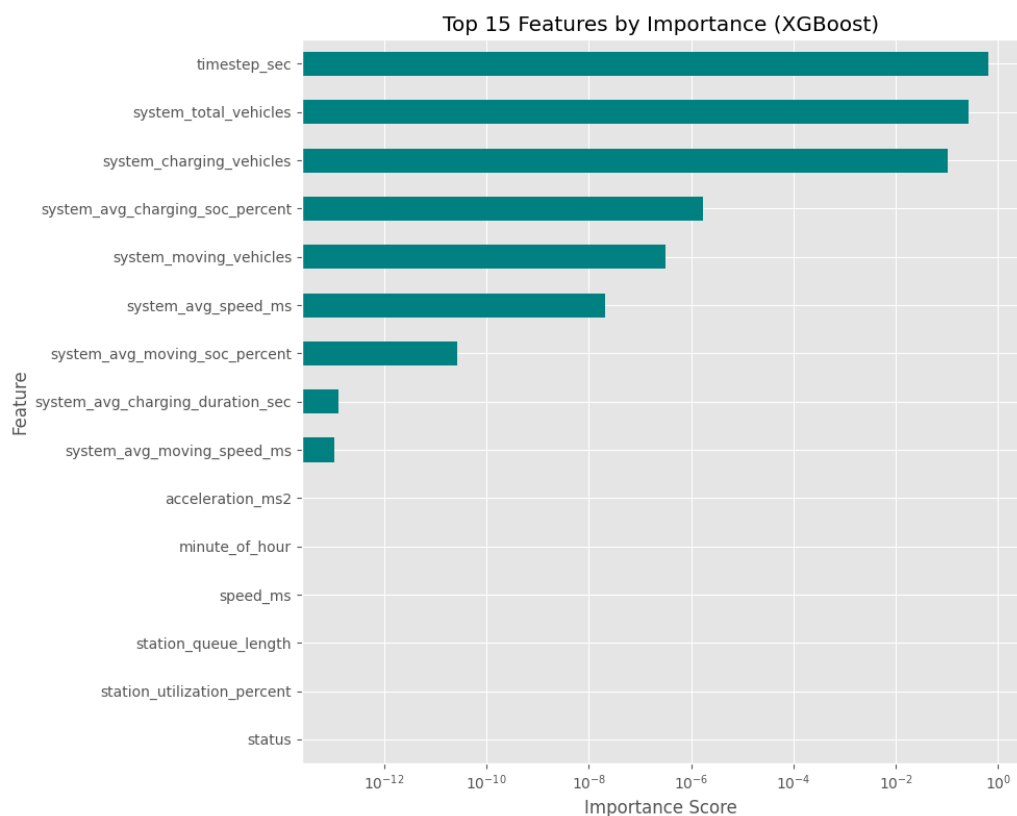


Figure 4.3: Top 15 features by importance according to the XGBoost model.

As shown in Figure 4.3, only a few dominant features—primarily `system_charging_vehicles`

and timestep_sec—drive the majority of the prediction accuracy. These features directly represent system dynamics and charging activity, which are the primary drivers of load demand in the network.

4.4.3 Correlation Analysis

Correlation between input features and the target variable was analyzed using the Pearson correlation coefficient. The heatmap in Figure 4.4 illustrates the strong positive correlation between vehicle status and system-wide load metrics.

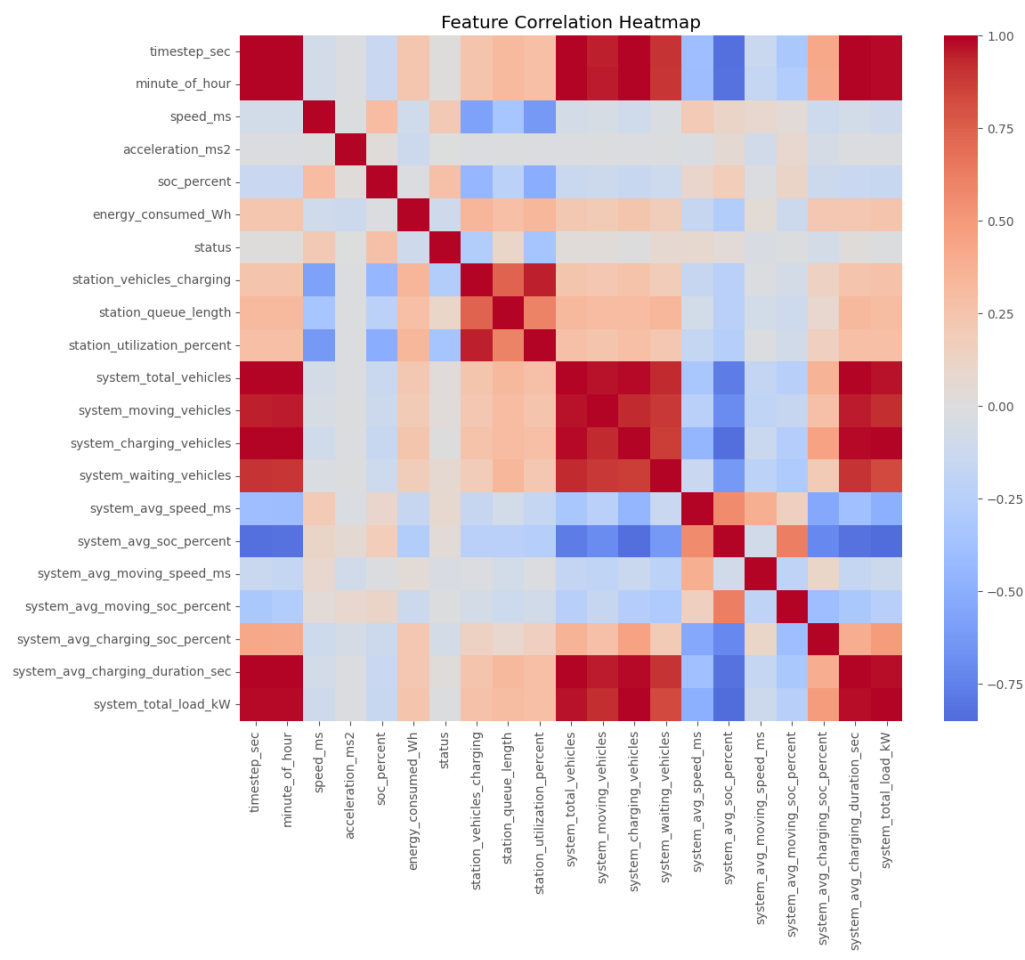


Figure 4.4: Heatmap of feature correlations in the EV charging dataset.

The bar chart in Figure 4.5 further confirms that load demand is primarily influenced by aggregated system behavior, such as the total number of vehicles charging, rather than individual

vehicle-level attributes like speed or acceleration.

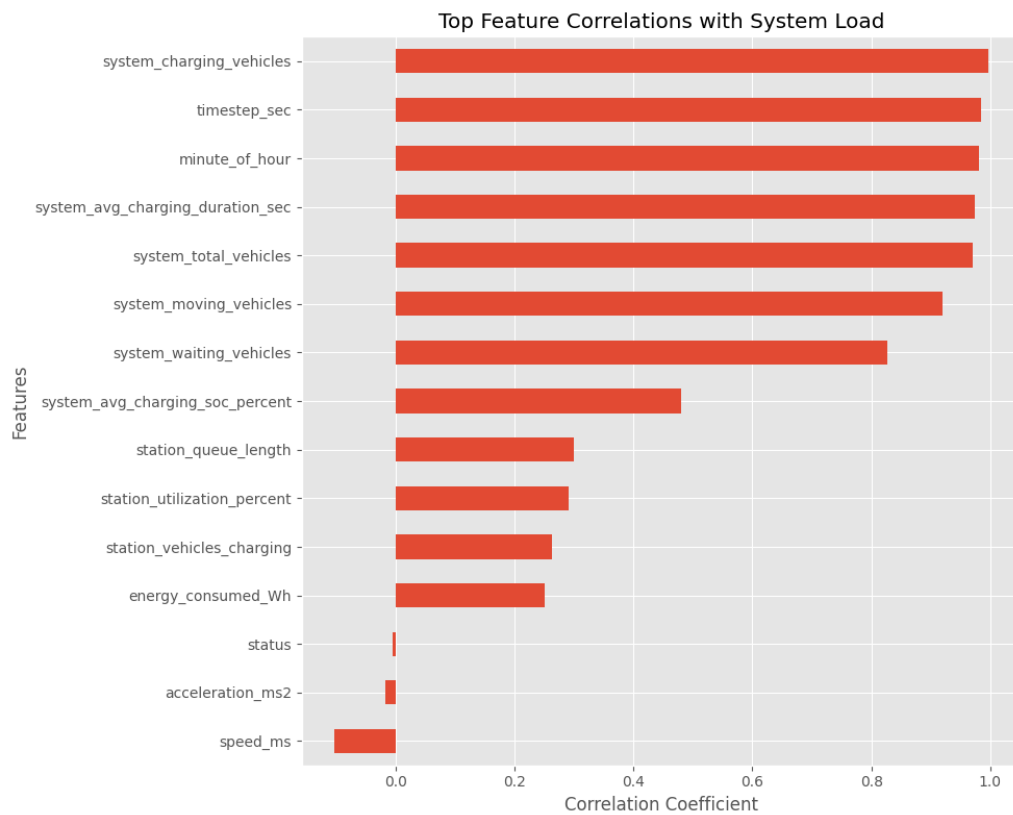


Figure 4.5: Top feature correlations with the system total load (kW).

4.4.4 Model Performance Justification

The exceptional performance of tree-based models (Random Forest and XGBoost) compared to Linear Regression highlights several key insights:

- **Tabular Suitability:** Tree models naturally handle tabular data and capture non-linear feature interactions without requiring explicit basis functions.
- **State-Driven Dynamics:** The prediction problem is inherently state-driven rather than sequence-driven. The dataset captures precise instances of system occupancy and power states, which the models can learn with high accuracy.
- **Handling Discontinuities:** Unlike Linear Regression, tree models can efficiently capture

the discontinuities in power demand caused by vehicles starting or stopping their charging sessions.

4.4.5 Analysis of Feature Contributions and Redundancy

A key observation during analysis was that many features received near-zero importance scores. This is primarily explained by two factors:

1. **Feature Redundancy:** Many variables (e.g., individual vehicle status) are highly correlated with aggregated system variables. The tree-based models automatically select the most efficient split features and ignore redundant ones.
2. **Aggregated Effects:** Effects such as vehicle speed and SOC are already implicitly represented in the "charging vs. driving" state, which is the stronger predictor.

This indicates that the dataset is highly structured and contains highly informative system-level variables, making the prediction problem relatively deterministic.

4.5 Discussion of Results and Final Conclusion

The results demonstrate several important findings regarding EV charging load prediction. The simulation framework successfully generates a rich, ML-ready dataset that mirrors complex real-world dynamics. The near-perfect model performance ($R^2 \approx 1.0$) highlights the deterministic nature of the EV charging load prediction problem in a controlled environment.

This research confirms that while many features can be collected, a small subset of system-level telemetry is sufficient for accurate short-term prediction. Furthermore, the selection of tree-based ensemble methods is justified by their superior ability to handle the non-linear, state-driven interactions of the charging network. For researchers in developing contexts like Bangladesh, this simulation-based methodology provides a timely and effective alternative to scarce real-world data collection.

4.6 Implications

The findings have several practical implications for smart grid and EV infrastructure management [2, 17]. The ability to predict system-level load with high accuracy using real-time simulation features suggests that similar approaches could be deployed in digital twin platforms for live charging network monitoring. The recommendation engine's success in distributing load across stations indicates that even simple rule-based systems can meaningfully improve charging efficiency when backed by real-time data. For CSE researchers at JUST and similar institutions in developing countries, the SUMO-based pipeline provides an accessible and cost-effective alternative to real-world data collection.

Chapter 5

Conclusion

Chapter 5: Conclusion

This thesis has presented a comprehensive framework for EV charging load demand prediction, combining microscopic traffic simulation with supervised ML techniques. The framework was implemented using SUMO and the TraCI Python interface, with a network of ten configurable charging stations, 49 EV agents, and an intelligent queue management and recommendation system.

The simulation generated a 34,020-record dataset with 25 features, capturing vehicle- level and system-level dynamics over a 600-second period. Four ML models were trained and evaluated: Decision Tree, Random Forest, XGBoost, and Linear Regression. Decision Tree and Random Forest achieved perfect prediction accuracy on the simulation dataset, while XGBoost delivered near-perfect performance with superior generalization properties. Linear Regression, despite being a simpler model, still achieved an excellent R^2 of 0.9992, highlighting the near-linear nature of the aggregate load signal.

5.1 Concluding Remarks

The key contributions of this research are:

1. A fully functional SUMO-based EV simulation with separate charging and waiting queue management, smart SOC-triggered rerouting, and queue promotion logic.
2. A rich, ML-ready dataset (`EV_Charging_Load_Demand_Dataset.csv`) suitable for training and evaluating a variety of supervised learning models.
3. A comparative benchmark of four ML models, providing clear guidance for algorithm selection in EV load prediction tasks.
4. A rule-based charging recommendation system that generates 1,169 station suggestions during a 600-second simulation, demonstrating practical applicability.

The results confirm that simulation-based data generation is a viable approach for addressing the scarcity of real-world EV charging datasets, particularly in developing-country contexts

such as Bangladesh where large-scale EV deployments are still nascent [10].

5.2 Limitations

Several limitations should be noted. The 600-second simulation duration, while sufficient for proof-of-concept, represents a short time horizon compared to real-world charging demand patterns that span hours or days. Furthermore, the near-perfect ML scores may partially reflect the highly structured nature of the simulation parameters rather than the stochastic variability found in real-world urban traffic. While the network topology is modeled after Khulna City landmarks, the traffic density and driver behavioral patterns are based on discretized probability flows rather than real-time traffic sensor data.

5.3 Recommendations for Further Research

Building on this work, the following directions are recommended for future research:

1. **Larger and more diverse simulations:** Extend the simulation to longer time horizons (24+ hours), larger vehicle fleets, and heterogeneous EV types to test model robustness [4].
2. **Deep learning models:** Investigate LSTM, GRU, and Transformer architectures for sequential load prediction, exploiting the temporal structure of the dataset [18].
3. **Real-world validation:** Validate the simulation-trained models against real-world EV charging data when such data becomes available in Bangladesh [10].
4. **Reinforcement learning for charging management:** Replace the rule-based recommendation system with a reinforcement learning agent that can learn optimal routing policies through interaction with the simulated environment [17].

5. **Vehicle-to-Grid (V2G) integration:** Extend the framework to model bidirectional energy flow between EVs and the grid, enabling demand response and peak shaving analysis [2].
6. **Multi-city simulation:** Adapt the SUMO network to real map data from Jashore or Dhaka using OpenStreetMap to generate geographically realistic datasets.

[19]

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